
Experiment 3

Building Y-Bus and Newton–Raphson Jacobian in MATLAB

Lab Report

Student Name HAMZA AHMAD

Registration Number 2022-EE-165

Section B

Power System Operation and Control Laboratory

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MATLAB code for Y-bus and Jacobian functions

```
%% ===== SYSTEM DATA =====  
  
BMva = 100;    % Base MVA  
  
% Bus Data Format:  
% Columns: Bus | Vsp | delta | Pg | Qg | Pl | Ql | Qmin | Qmax | Type  
% Bus Type: 1=Slack, 2=PV (Generator), 3=PQ (Load)  
  
busdata = [  
    1   1.060   0   232.4   -16.9   0.0   0.0   0   0   1;  
    2   1.045   0   40.0   43.56  21.7  12.7  -40  50  2;  
    3   1.010   0    0.0    0.0  94.2  19.0  -25  40  3;  
    4   1.000   0    0.0    0.0  47.8  -3.9   0   0  3;  
    5   1.000   0    0.0    0.0   7.6   1.6   0   0  3;  
    6   1.070   0    0.0    0.0  11.2   7.5   -6  24  2;
```

```

7   1.000  0      0.0    0.0    0.0    0.0    0    0    3;
8   1.090  0      0.0    0.0    0.0    0.0    0    0    2;
9   1.000  0      0.0    0.0    29.5   16.6    0    0    3;
10  1.000  0      0.0    0.0    9.0    5.8    0    0    3;
11  1.000  0      0.0    0.0    3.5    1.8    0    0    3;
12  1.000  0      0.0    0.0    6.1    1.6    0    0    3;
13  1.000  0      0.0    0.0    13.5   5.8    0    0    3;
14  1.000  0      0.0    0.0    14.9   5.0    0    0    3;

```

```
];
```

```
% Line Data Format:
```

```
% Columns: From | To | R | X | B/2 | Tap
```

```
linedata = [
```

```

1   2   0.01938   0.05917   0.0264   1.000;
1   5   0.05403   0.22304   0.0246   1.000;
2   3   0.04699   0.19797   0.0219   1.000;
2   4   0.05811   0.17632   0.0170   1.000;
2   5   0.05695   0.17388   0.0173   1.000;
3   4   0.06701   0.17103   0.0064   1.000;
4   5   0.01335   0.04211   0.0000   1.000;
4   7   0.00000   0.20912   0.0000   0.978;   % Transformer
4   9   0.00000   0.55618   0.0000   0.969;   % Transformer
5   6   0.00000   0.25202   0.0000   1.000;
6  11   0.09498   0.19890   0.0000   1.000;
6  12   0.12291   0.25581   0.0000   1.000;
6  13   0.06615   0.13027   0.0000   1.000;
7   8   0.00000   0.17615   0.0000   1.000;
7   9   0.00000   0.11001   0.0000   1.000;
9  10   0.03181   0.08450   0.0000   1.000;
9  14   0.12711   0.27038   0.0000   1.000;
10  11   0.08205   0.19207   0.0000   1.000;
12  13   0.22092   0.19988   0.0000   1.000;
13  14   0.17093   0.34802   0.0000   1.000;

```

```
];
```

```

%% ===== Y-BUS FORMATION =====

fprintf('Building Y-bus matrix...\n');

nbus = max(max(linedata(:, 1:2)));

Y = zeros(nbus, nbus);

fb = linedata(:, 1); % From bus
tb = linedata(:, 2); % To bus

for k = 1:size(linedata, 1)
    i = fb(k);
    j = tb(k);

    R = linedata(k, 3);
    X = linedata(k, 4);
    Bc = linedata(k, 5);
    tap = linedata(k, 6);

    if tap == 0
        tap = 1;
    end

    Z = R + 1i * X;
    y = 1 / Z;
    ysh = 1i * Bc;

    % Off-diagonal elements
    Y(i, j) = Y(i, j) - y / tap;
    Y(j, i) = Y(i, j);

    % Diagonal elements
    Y(i, i) = Y(i, i) + y / (tap^2) + ysh;
    Y(j, j) = Y(j, j) + y + ysh;
end

G = real(Y); % Conductance matrix
B = imag(Y); % Susceptance matrix

%% ===== DISPLAY Y-BUS MATRIX =====

```

```

fprintf('\n=====\\n');
fprintf('  Y-BUS MATRIX (ADMITTANCE MATRIX)\\n');
fprintf('=====\\n\\n');
fprintf('Y-BUS in Rectangular Form (Real + j*Imag):\\n\\n');
for i = 1:nbus
    fprintf('Bus %2d: ', i);
    for j = 1:nbus
        if abs(Y(i,j)) > 1e-6
            fprintf(['%7.4f%+7.4fj] ', real(Y(i,j)), imag(Y(i,j))]);
        else
            fprintf('[ 0.0000 +0.0000j] ');
        end
    end
    fprintf('\\n');
end
fprintf('\\n\\nCONDUCTANCE MATRIX (G = Real part of Y):\\n\\n');
for i = 1:nbus
    fprintf('Bus %2d: ', i);
    for j = 1:nbus
        fprintf('%8.4f ', G(i,j));
    end
    fprintf('\\n');
end
fprintf('\\n\\nSUSCEPTANCE MATRIX (B = Imaginary part of Y):\\n\\n');
for i = 1:nbus
    fprintf('Bus %2d: ', i);
    for j = 1:nbus
        fprintf('%8.4f ', B(i,j));
    end
    fprintf('\\n');
end
fprintf('\\n=====\\n\\n');

```

```
% ===== JACOBIAN MATRIX CONSTRUCTION =====
```

```
% J1: dP/d(delta)
```

```
J1 = zeros(nbus - 1);
```

```
for i = 1:nbus - 1
```

```
    m = i + 1;
```

```
    for k = 1:nbus - 1
```

```
        n = k + 1;
```

```
        if n == m
```

```
            for n2 = 1:nbus
```

```
                J1(i, k) = J1(i, k) + V(m) * V(n2) * ...
```

```
                    (-G(m,n2) * sin(del(m) - del(n2)) + ...
```

```
                    B(m,n2) * cos(del(m) - del(n2)));
```

```
            end
```

```
            J1(i, k) = J1(i, k) - V(m)^2 * B(m, m);
```

```
        else
```

```
            J1(i, k) = V(m) * V(n) * ...
```

```
                (G(m,n) * sin(del(m) - del(n)) - ...
```

```
                B(m,n) * cos(del(m) - del(n)));
```

```
        end
```

```
    end
```

```
end
```

```
% J2: dP/dV
```

```
J2 = zeros(nbus - 1, npq);
```

```
for i = 1:nbus - 1
```

```
    m = i + 1;
```

```
    for k = 1:npq
```

```
        n = pq(k);
```

```
        if n == m
```

```
            for n2 = 1:nbus
```

```
                J2(i, k) = J2(i, k) + V(n2) * ...
```

```

        (G(m,n2) * cos(del(m) - del(n2)) + ...
        B(m,n2) * sin(del(m) - del(n2)));

    end

    J2(i, k) = J2(i, k) + V(m) * G(m, m);

else

    J2(i, k) = V(m) * ...

        (G(m,n) * cos(del(m) - del(n)) + ...
        B(m,n) * sin(del(m) - del(n)));

end

end

end

end

% J3: dQ/d(delta)
J3 = zeros(npq, nbus - 1);

for i = 1:npq
    m = pq(i);

    for k = 1:nbus - 1
        n = k + 1;

        if n == m

            for n2 = 1:nbus

                J3(i, k) = J3(i, k) + V(m) * V(n2) * ...

                    (G(m,n2) * cos(del(m) - del(n2)) + ...
                    B(m,n2) * sin(del(m) - del(n2)));

            end

            J3(i, k) = J3(i, k) - V(m)^2 * G(m, m);

        else

            J3(i, k) = V(m) * V(n) * ...

                (-G(m,n) * cos(del(m) - del(n)) - ...
                B(m,n) * sin(del(m) - del(n)));

        end

    end

end

end

end

```

```

% J4: dQ/dV

J4 = zeros(npq);

for i = 1:npq
    m = pq(i);

    for k = 1:npq
        n = pq(k);

        if n == m

            for n2 = 1:nbus

                J4(i, k) = J4(i, k) + V(n2) * ...

                    (G(m,n2) * sin(del(m) - del(n2)) - ...

                    B(m,n2) * cos(del(m) - del(n2)));

            end

            J4(i, k) = J4(i, k) - V(m) * B(m, m);

        else

            J4(i, k) = V(m) * ...

                (G(m,n) * sin(del(m) - del(n)) - ...

                B(m,n) * cos(del(m) - del(n)));

        end

    end

end

end

% ===== DISPLAY JACOBIAN MATRICES (FIRST ITERATION) =====

if display_jacobian

    fprintf('\n===== \n');

    fprintf('  JACOBIAN MATRICES (Iteration 1) \n');

    fprintf('===== \n \n');

    fprintf('J1 Matrix (dP/d_delta) [%dx%d]: \n', size(J1,1),
size(J1,2));

    fprintf('(Partial derivatives of real power w.r.t. voltage
angles) \n \n');

    disp(J1);

    fprintf('\nJ2 Matrix (dP/dV) [%dx%d]: \n', size(J2,1), size(J2,2));

```

```

        fprintf(' (Partial derivatives of real power w.r.t. voltage
magnitudes)\n\n');

        disp(J2);

        fprintf('\nJ3 Matrix (dQ/d_delta) [%dx%d]:\n', size(J3,1),
size(J3,2));

        fprintf(' (Partial derivatives of reactive power w.r.t. voltage
angles)\n\n');

        disp(J3);

        fprintf('\nJ4 Matrix (dQ/dV) [%dx%d]:\n', size(J4,1), size(J4,2));

        fprintf(' (Partial derivatives of reactive power w.r.t. voltage
magnitudes)\n\n');

        disp(J4);

        fprintf('=====\n\n');

        display_jacobian = false;

end

% Assemble full Jacobian

J = [J1 J2; J3 J4];

```

RESULTS

Y-bus and Jacobian Matrices

Y-BUS in Rectangular Form (Real + j*Imag):

```
Bus 1: [ 6.0250-19.4471j] [-4.9991+15.2631j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.0259+4.2350j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +
Bus 2: [-4.9991+15.2631j] [ 9.5213-30.2721j] [-1.1350+4.7819j] [-1.6860+5.1158j] [-1.7011+5.1939j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.00
Bus 3: [ 0.0000 +0.0000j] [-1.1350+4.7819j] [ 3.1210-9.8224j] [-1.9860+5.0688j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.
Bus 4: [ 0.0000 +0.0000j] [-1.6860+5.1158j] [-1.9860+5.0688j] [10.5130-38.6542j] [-6.8410+21.5786j] [ 0.0000 +0.0000j] [ 0.0000+4.8895j] [ 0.0000 +0.00
Bus 5: [-1.0259+4.2350j] [-1.7011+5.1939j] [ 0.0000 +0.0000j] [-6.8410+21.5786j] [ 9.5680-34.9335j] [ 0.0000+3.9679j] [ 0.0000 +0.0000j] [ 0.0000 +0.00
Bus 6: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000+3.9679j] [ 6.5799-17.3407j] [ 0.0000 +0.0000j] [ 0.0000
Bus 7: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000+4.8895j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000-19.5490j] [ 0.0000+5
Bus 8: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000+5.6770j] [ 0.0000-
Bus 9: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000+4.8895j] [ 0.0000 +0.0000j] [ 0.0000+9.0901j] [ 0.0000 +
Bus 10: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.00
Bus 11: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.9550+4.0941j] [ 0.0000 +0.0000j] [ 0.0000
Bus 12: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.5260+3.1760j] [ 0.0000 +0.0000j] [ 0.0000
Bus 13: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-3.0989+6.1028j] [ 0.0000 +0.0000j] [ 0.0000
Bus 14: [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.00
```

```
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
+0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000+1.8555j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
+0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.9550+4.0941j] [-1.5260+3.1760j] [-3.0989+6.1028j] [ 0.0000 +0.0000j]
.0000-19.5490j] [ 0.0000+5.6770j] [ 0.0000+9.0901j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000+5.6770j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
.0000+9.0901j] [ 0.0000 +0.0000j] [ 5.3261-24.2825j] [-3.9020+10.3654j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.4240+3.0291j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-3.9020+10.3654j] [ 5.7829-14.7683j] [-1.8809+4.4029j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.8809+4.4029j] [ 3.8359-8.4970j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 4.0150-5.4279j] [-2.4890+2.2520j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-2.4890+2.2520j] [ 6.7249-10.6697j] [-1.1370+2.3150j]
0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.4240+3.0291j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [ 0.0000 +0.0000j] [-1.1370+2.3150j] [ 2.5610-5.3440j]
```

J1 Matrix (dP/d_delta) [13x13]:
(Partial derivatives of real power w.r.t. voltage angles)

32.7276	-5.0470	-5.3461	-5.4277	0	0	0	0	0	0	0	0	0	0
-5.0470	10.1665	-5.1195	0	0	0	0	0	0	0	0	0	0	0
-5.3461	-5.1195	38.7891	-21.5786	0	-4.8895	0	-1.8555	0	0	0	0	0	0
-5.4277	0	-21.5786	35.7410	-4.2457	0	0	0	0	0	0	0	0	0
0	0	0	-4.2457	18.5546	0	0	0	0	-4.3807	-3.3983	-6.5299	0	0
0	0	-4.8895	0	0	20.1675	-6.1879	-9.0901	0	0	0	0	0	0
0	0	0	0	0	-6.1879	6.1879	0	0	0	0	0	0	0
0	0	-1.8555	0	0	-9.0901	0	24.3400	-10.3654	0	0	0	-3.0291	0
0	0	0	0	0	0	0	-10.3654	14.7683	-4.4029	0	0	0	0
0	0	0	0	-4.3807	0	0	0	-4.4029	8.7836	0	0	0	0
0	0	0	0	-3.3983	0	0	0	0	0	5.6503	-2.2520	0	0
0	0	0	0	-6.5299	0	0	0	0	0	-2.2520	11.0969	-2.3150	0
0	0	0	0	0	0	0	-3.0291	0	0	0	-2.3150	5.3440	0

J2 Matrix (dP/dV) [13x10]:
(Partial derivatives of real power w.r.t. voltage magnitudes)

-1.1861	-1.7619	-1.7777	0	0	0	0	0	0	0
3.1323	-2.0058	0	0	0	0	0	0	0	0
-1.9860	10.4173	-6.8410	0	0	0	0	0	0	0
0	-6.8410	9.4299	0	0	0	0	0	0	0
0	0	0	0	0	0	-2.0919	-1.6328	-3.3159	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	5.3261	-3.9020	0	0	0	-1.4240
0	0	0	0	-3.9020	5.7829	-1.8809	0	0	0
0	0	0	0	0	-1.8809	3.6991	0	0	0
0	0	0	0	0	0	0	3.9082	-2.4890	0
0	0	0	0	0	0	0	-2.4890	6.5080	-1.1370
0	0	0	0	-1.4240	0	0	0	-1.1370	2.5610

J3 Matrix (dQ/d_delta) [10x13]:
(Partial derivatives of reactive power w.r.t. voltage angles)

1.1980	-3.2038	2.0058	0	0	0	0	0	0	0	0	0	0
1.7619	2.0058	-10.6087	6.8410	0	0	0	0	0	0	0	0	0
1.7777	0	6.8410	-9.7061	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	-5.3261	3.9020	0	0	0	1.4240
0	0	0	0	0	0	0	3.9020	-5.7829	1.8809	0	0	0
0	0	0	0	2.0919	0	0	0	1.8809	-3.9728	0	0	0
0	0	0	0	1.6328	0	0	0	0	0	-4.1218	2.4890	0
0	0	0	0	3.3159	0	0	0	0	0	2.4890	-6.9419	1.1370
0	0	0	0	0	0	0	1.4240	0	0	0	1.1370	-2.5610

J4 Matrix (dQ/dV) [10x10]:
(Partial derivatives of reactive power w.r.t. voltage magnitudes)

9.7753	-5.1195	0	0	0	0	0	0	0	0
-5.0688	38.5192	-21.5786	-4.8895	-1.8555	0	0	0	0	0
0	-21.5786	34.1260	0	0	0	0	0	0	0
0	-4.8895	0	18.9305	-9.0901	0	0	0	0	0
0	-1.8555	0	-9.0901	24.2250	-10.3654	0	0	0	-3.0291
0	0	0	0	-10.3654	14.7683	-4.4029	0	0	0
0	0	0	0	0	-4.4029	8.2104	0	0	0
0	0	0	0	0	0	0	5.2056	-2.2520	0
0	0	0	0	0	0	0	-2.2520	10.2425	-2.3150
0	0	0	0	-3.0291	0	0	0	-2.3150	5.3440

=====